

Generation Interconnection System Impact Study Report

For Queue Project Q19-1201

Clover 138 kV

June 27, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1201. The Interconnection Customer proposes a 79.2 MW wind facility interconnecting at the Clover 138 kV substation in the Virginia area. The request was received on May 04, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 440 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 8 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$66,568,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
DEWALT / OLD HICKORY 138.0 kV	BEAR / CLOVER 345.0 kV	157.2	275	5.9
DEWALT / GENEVA BORDER CITY 138.0 kV	BEAR / CLOVER 345.0 kV	145.9	306	5.9
GENEVA BORDER CITY / KEOKUK 138.0 kV L	KEOKUK / 32ND ST 138.0 kV	144.7	193	8.7
KEOKUK / 32ND ST 138.0 kV	KEOKUK / GENEVA BORDER CITY 138.0 kV line	115.1	237	6.7
STRAITS / 32ND ST SUBSTATION 138.0 kV	KEOKUK / GENEVA BORDER CITY 138.0 kV	113.6	236	6.7
DPC ELROY / STRAITS 138.0 kV	KEOKUK / GENEVA BORDER CITY 138.0 kV line	112.6	234	6.7
BEAR / MECKLENBURG POWER STA 345.0 kV	BEAR / CLOVER 345.0 kV	100.8	2,827	27.5
BEAR / FLUVANNA P.S 345.0 kV	BEAR / LEAVITT 345.0 kV line	100.7	1,358	22.4

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
DEWALT / OLD HICKORY 138.0 kV	Rebuild line with higher-capacity conductor	\$4,337,000	100% (\$4,337,000)
DEWALT / GENEVA BORDER CITY 138.0 kV	Rebuild line with higher-capacity conductor	\$20,635,000	100% (\$20,635,000)
GENEVA BORDER CITY / KEOKUK 138.0 kV	Rebuild line with higher-capacity conductor	\$5,691,000	100% (\$5,691,000)
KEOKUK / 32ND ST 138.0 kV	Rebuild line with higher-capacity conductor	\$5,780,000	100% (\$5,780,000)
STRAITS / 32ND ST SUBSTATION 138.0 kV	Rebuild line with higher-capacity conductor	\$12,829,000	100% (\$12,829,000)
DPC ELROY / STRAITS 138.0 kV	Rebuild line with higher-capacity conductor	\$15,743,000	100% (\$15,743,000)
BEAR / MECKLENBURG POWER STA 345.0 kV	Reconductor line	\$34,934,000	2% (\$593,000)
BEAR / FLUVANNA P.S SUBSTATION 345.0 kV	Reconductor line	\$61,069,000	2% (\$960,000)
	Total allocated to project		\$66,568,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.